

4(a)	(strain =) extension / <u>original</u> length	B1
4(b)(i)	$E = \sigma / \varepsilon$	C1
	maximum stress = $2.1 \times 10^{11} \times 4.0 \times 10^{-4}$ $= 8.4 \times 10^7 \text{ Pa}$	A1
4(b)(ii)	$\sigma = F/A$	C1
	minimum area = $8.0 \times 10^3 / 8.4 \times 10^7$ $= 9.5 \times 10^{-5} \text{ m}^2$	A1

Question	Answer	Marks
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